Pollinator Network Draft and Build

Design a habitat that works all season

At a glance: Draft a pollinator habitat that works all season, not just on the prettiest day. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: plant-pollinator networks, seasonal coverage, habitat design, systems thinking (Understand).
- Apply design constraints (native plants, clumping, and full-season bloom) to build a pollinator network and justify the trade-offs (Apply).
- Use their layout to judge whether a redesign supports more pollinators across the season (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- plant card decks: one per team (4, plus 1 spare)
- pollinator card decks: one per team, in the same deck set (4, plus 1 spare)
- grid boards: one per team (4, plus 1 spare)
- season tokens: a spring, summer, fall set per team (about 80 total)
- markers: shared, 1 to 2 per team
- stickers: shared, about 1 sheet per team

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Stage the scoring sheet and any reference values before testing begins.

03 No PPE is needed: this is a dry card-and-grid game with no insects, soil, water, or chemicals. Keep small tokens and stickers accounted for.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Draft a pollinator habitat that works all season, not just on the prettiest day." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: native and clumping logic, cover spring, summer, and fall, and change one placement at a time, rechecking coverage before rearranging more. Follow the safety notes.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask how their layout covers every season and why each clump helps, and resist solving it for them.

Phase 5 Build the network

Teams lay plants and pollinators on the grid, place season tokens, and check that spring, summer, and fall are each covered.

Phase 6 Refine

Each team finds its weakest season or an unfed pollinator, adjusts the layout (add a clump, swap in a native, fill a season gap), and rechecks coverage.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Return all cards, tokens, and boards to the bin so the next team has a complete set; no PPE and nothing to wash.

TROUBLESHOOTING

Problem: A team rearranges several placements at once. **Fix:** Have them change one rule at a time (lock a clump or fill one open season) and check it against the rubric before changing more.

Problem: A team leaves a season uncovered. Fix: Point to the empty season token slot and ask which plant could bloom then; have them draft a plant for the missing season before scoring.

Problem: A team maps each flower to just one pollinator. **Fix:** Most flowers feed several kinds of visitors, so accept any plausible match the card supports and reward full-season coverage over a single keyed answer.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to defend how their network supports the most pollinators across the whole season.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to networks, not single plants.

Tier 2 (building but not reasoning):

- "What one change to your layout would help the most pollinators, and why?"

Tier 3 (ready to extend):

- "Design a second version with a different placement strategy and argue which one supports more pollinators across the season."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Where was the biggest gap in your bloom calendar?
- Why does clumping help pollinators?
- What happens to the network if one season is missing?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Seasonal coverage	30
Pollinator fit	25
Native and clumping logic	20
Layout clarity	15
Defense	10
Total	**100**

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES USDA Pollinators; US Forest Service pollinator guidance

Plant cards

Each plant is native here. The dark season chips show when it blooms; the visitors are the pollinators it feeds. Place plants so something blooms in spring, summer, and fall, and group the same plant in clumps.

Serviceberry
Native

Amelanchier canadensis

BLOOMS

SPRING SUMMER FALL

VISITED BY bees, flies

One of the first trees to bloom each April.

Golden Alexanders
Native

Zizia aurea

BLOOMS

SPRING **SUMMER** FALL

VISITED BY bees, flies, beetles, butterflies

Tiny yellow flowers feed early short-tongued insects.

Eastern Redbud
Native

Cercis canadensis

BLOOMS

SPRING SUMMER FALL

VISITED BY bees, butterflies

Pink flowers bloom on bare branches in April.

Butterfly Weed
Native

Asclepias tuberosa

BLOOMS

SPRING **SUMMER** FALL

VISITED BY butterflies, bees, hummingbirds

Orange milkweed and a monarch caterpillar host.

Wild Bergamot
Native

Monarda fistulosa

BLOOMS

SPRING **SUMMER** FALL

VISITED BY bees, butterflies, hummingbirds, moths

Lavender pompoms that almost everyone visits.

Black-eyed Susan
Native

Rudbeckia hirta

BLOOMS

SPRING **SUMMER** **FALL**

VISITED BY bees, butterflies, flies

Golden petals with a dark center; a summer classic.

Mountain Mint
Native

Pycnanthemum tenuifolium

BLOOMS

SPRING **SUMMER** **FALL**

VISITED BY bees, flies, butterflies, beetles

Flat white flowers draw a huge mix of insects.

Summersweet
Native

Clethra alnifolia

BLOOMS

SPRING **SUMMER** FALL

VISITED BY bees, butterflies, hummingbirds

Sweet-smelling white spikes for shady, wet spots.

Cardinal Flower
Native

Lobelia cardinalis

BLOOMS

SPRING **SUMMER** **FALL**

VISITED BY hummingbirds, butterflies

Red tubes built for hummingbird beaks.

Joe-Pye Weed
Native

Eutrochium purpureum

BLOOMS

SPRING **SUMMER** **FALL**

VISITED BY butterflies, bees

Tall pink clouds buzzing in late summer.

New England Aster
Native

Symphyotrichum novae-angliae

BLOOMS

SPRING SUMMER **FALL**

VISITED BY bees, butterflies, flies

Purple fall blooms fuel migrating monarchs.

Goldenrod
Native

Solidago rugosa

BLOOMS

SPRING SUMMER **FALL**

VISITED BY bees, butterflies, flies, beetles

Yellow plumes blooming right up to frost.

Pollinator cards

Each pollinator needs flowers in bloom across every season it is active. Match it to plants whose flower type fits (hummingbirds to red tubes, flies and beetles to flat open clusters, bees and butterflies to most flowers).

Native Bees

ACTIVE

SPRING SUMMER FALL

VISITS Open daisy, mint, and clustered flowers with easy pollen.

Most are solitary and do not sting.

Bumble Bees

ACTIVE

SPRING SUMMER FALL

VISITS Tube and hooded flowers like bergamot, reached with long tongues.

Queens wake first and need early spring blooms.

Butterflies and Monarchs

ACTIVE

SPRING SUMMER FALL

VISITS Flat-topped clusters and big landing-pad flowers.

Monarchs fuel on fall asters before migrating.

Hummingbirds

ACTIVE

SPRING SUMMER FALL

VISITS Red and pink tubular flowers full of nectar.

Long beaks reach where bees cannot.

Hoverflies and Flies

ACTIVE

SPRING SUMMER FALL

VISITS Shallow open flowers like mountain mint.

Hoverflies copy bees but cannot sting.

Beetles

ACTIVE

SPRING SUMMER FALL

VISITS Flat, sturdy flower clusters they can crawl across.

Some of Earth's oldest pollinators.

Grid board

Place your plant cards in the season they bloom (a plant that blooms twice can sit in two columns). Keep something in every column, and clump the same plant together.

SPRING	SUMMER	FALL

Pollinator food check

Tick a box only if a plant in that season feeds this pollinator. Every pollinator should have a tick in each season it is active.

POLLINATOR	SPRING	SUMMER	FALL
Native Bees			
Bumble Bees			
Butterflies and Monarchs			
Hummingbirds			
Hoverflies and Flies			
Beetles			